MASKER: Masked Keyword Regularization for Reliable Text Classification

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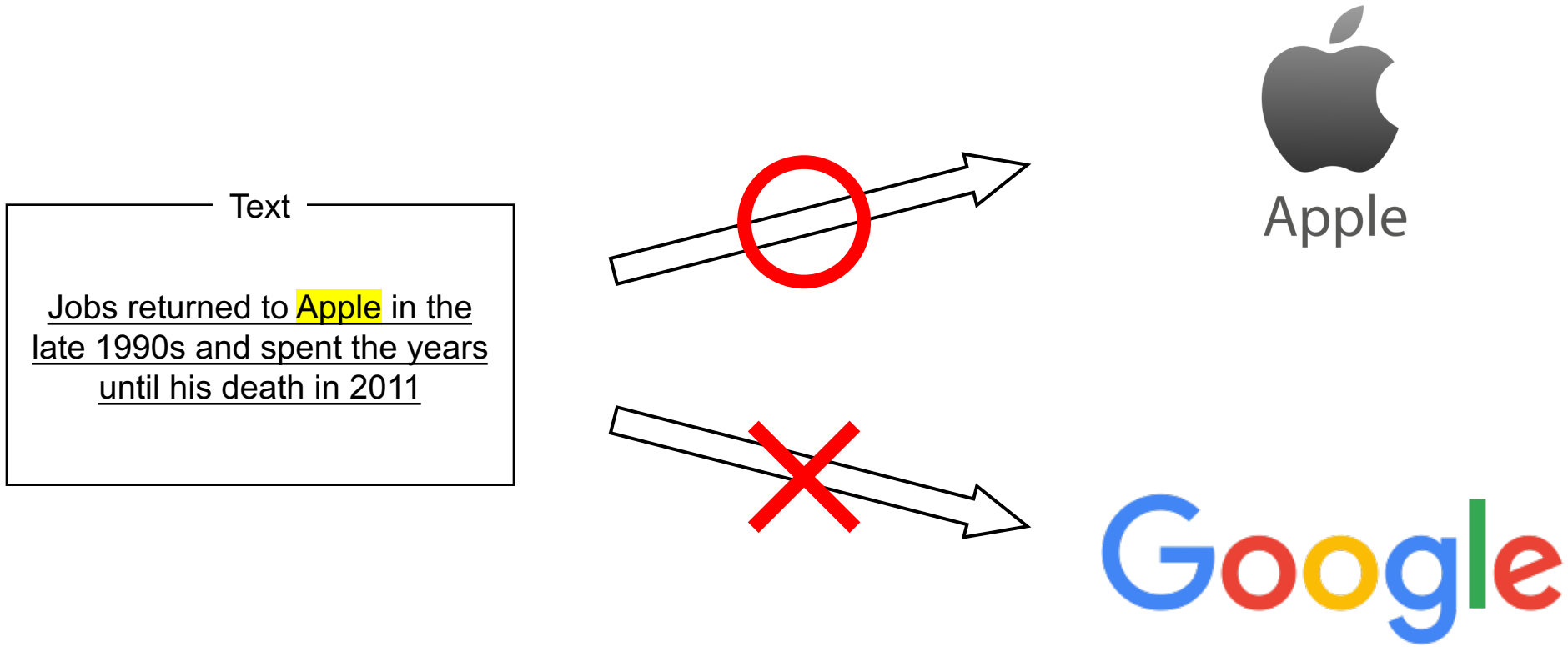
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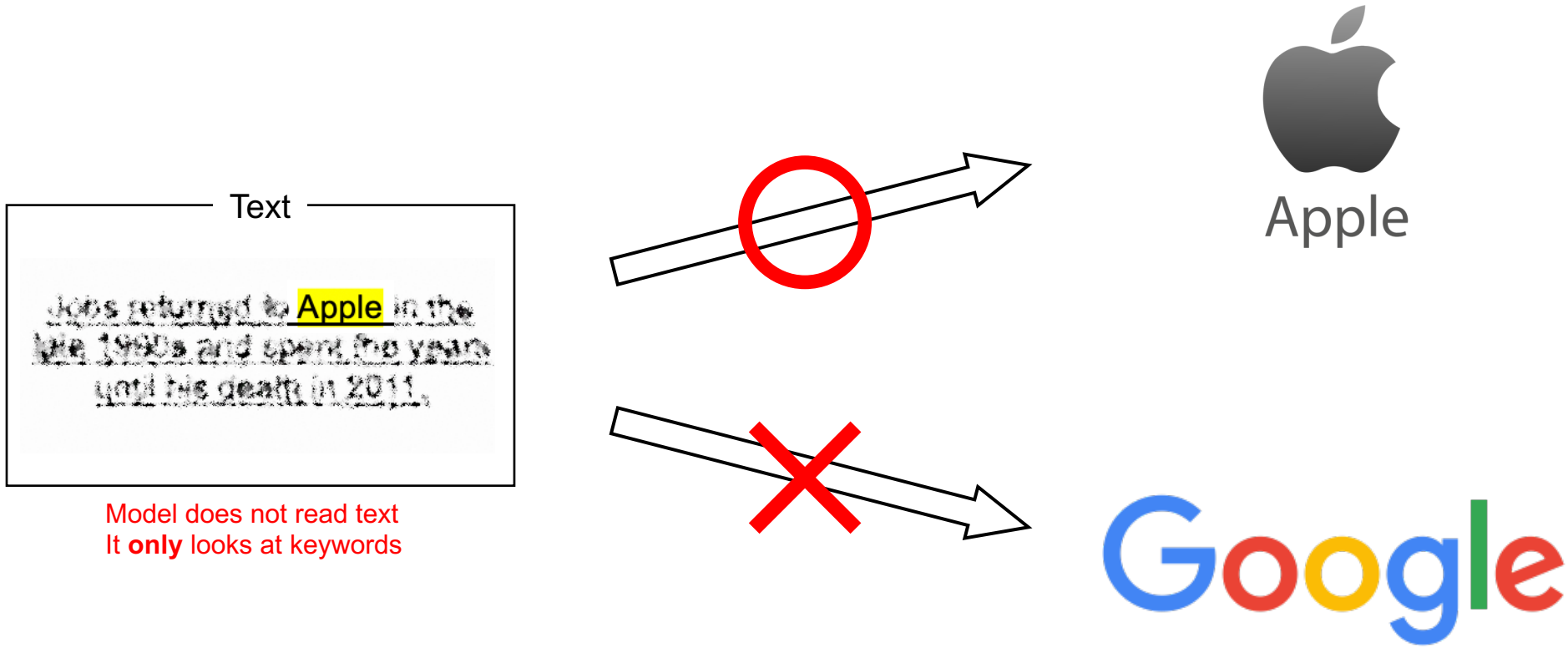
Language models over-rely on keywords

- We expect language models to **understand** and classify texts
  - As a human, language models should comprehend the context of input texts



Language models over-rely on keywords

- Models can find easy-ways for classification by **only** using keywords
  - They learn to classify texts **without** understanding the whole context



Language models over-rely on keywords

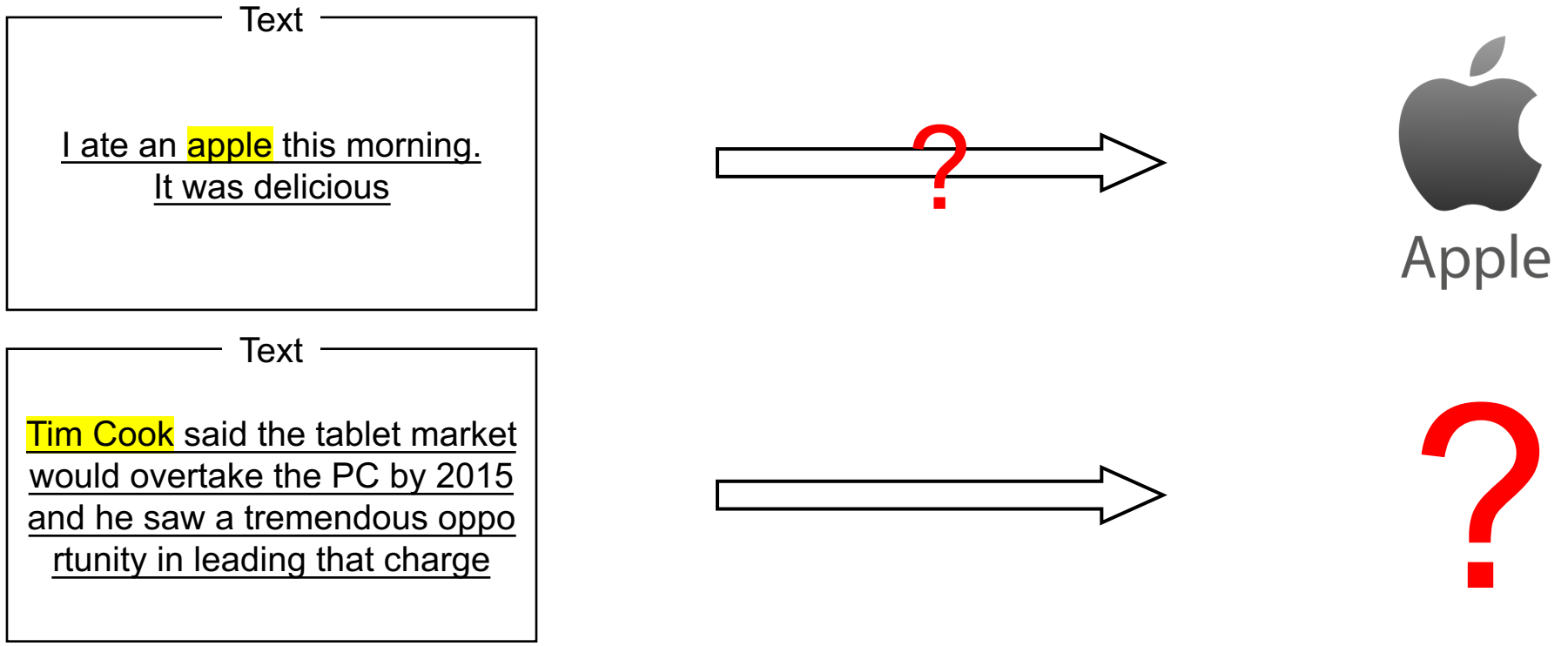
- Keyword obviously contains key information for classification

  It is natural to see keywords for classification

- But **why** over-reliance on keywords is problematic?

Out-of-distribution and domain generalization

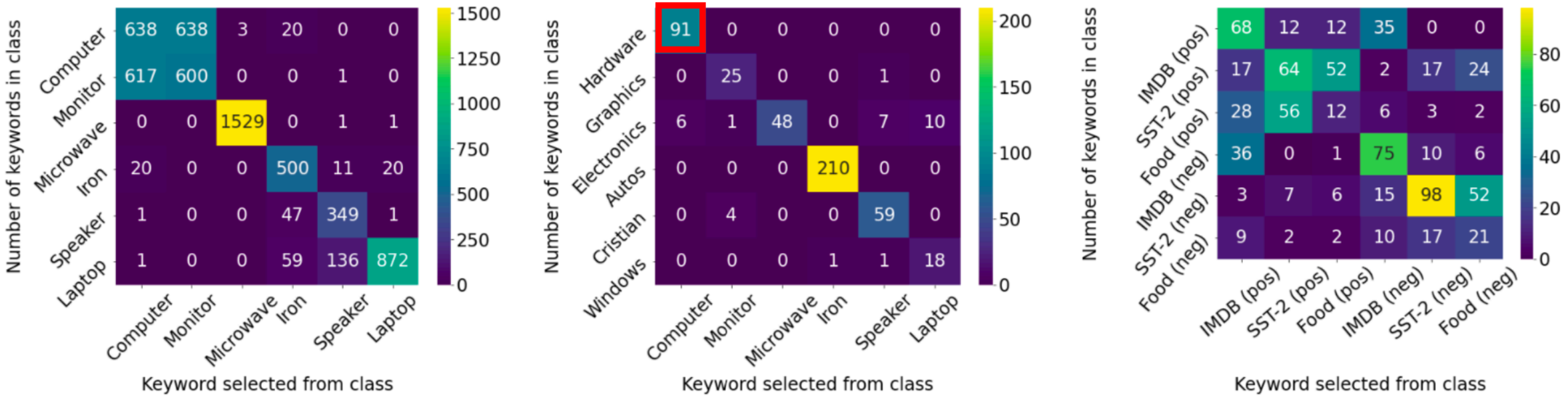
- Let's think about the following cases:
  - Out-of-distribution(OOD) data which contains keywords **IN** training set
  - Text from cross-domain which contains keywords, but which is **NOT IN** training set



Keywords overlaps in various dataset

- We pick keywords from the training set, and count the frequency of each keyword in
  - In-distribution (training) dataset, OOD dataset, Cross-domain dataset
- Indeed, overlap among different datasets, even if they are not closely related

A Keyword from class ‘Computer’ appears 91 times at OOD class ‘Hardware’



(a) In-distribution (b) Out-of-distribution (c) Cross-domain

Toward reliable text classification

- We should handle the over-reliance problem of NLP models!
- We focus on…
  - Training models to focus on the **context** around keywords
  - Training models not to classify texts **only with the existence** of specific keywords
  - **Finding keywords** that occur bias for classification

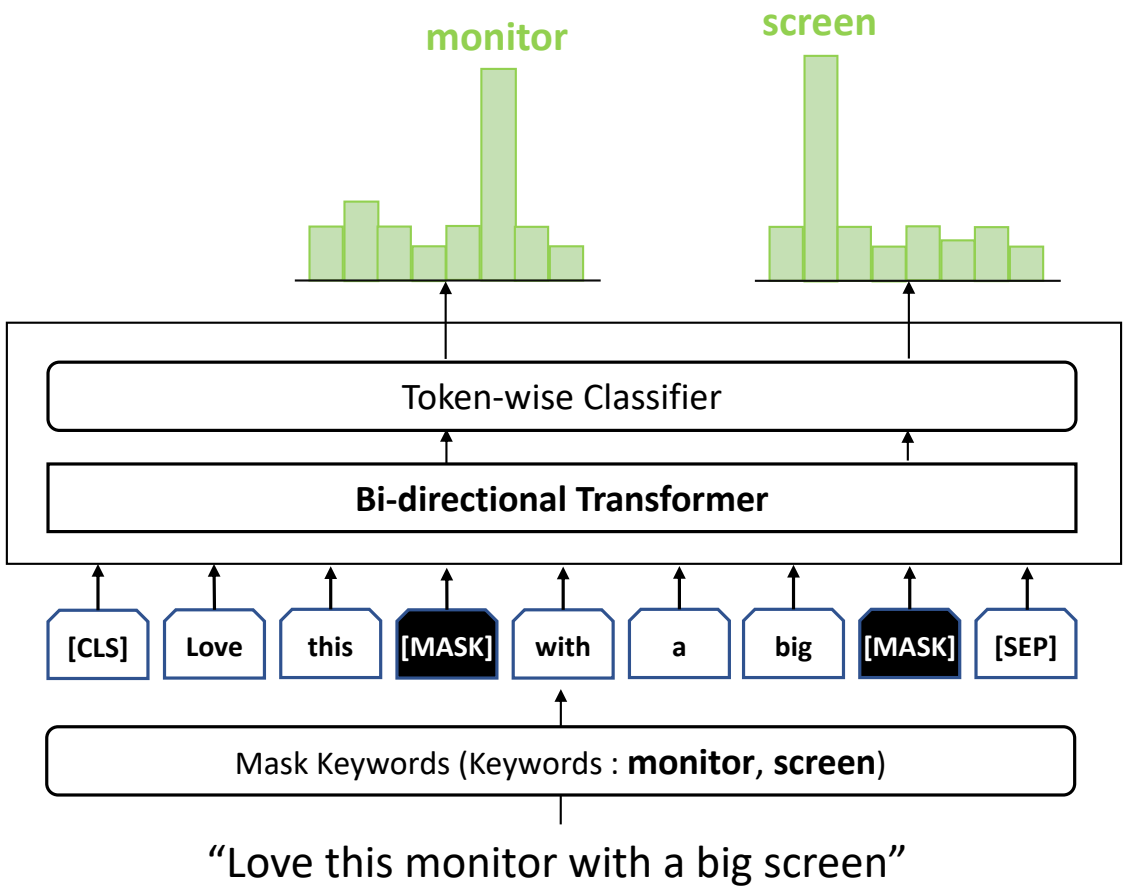
MKR: Following context around a few keywords

- Regularizing model to see the context around keywords would be effective
- MKR is inspired by Masked Language Modeling(MLM), a pre-training method for BERT

⇨ We figure out **Masked Keyword Reconstruction (MKR)**

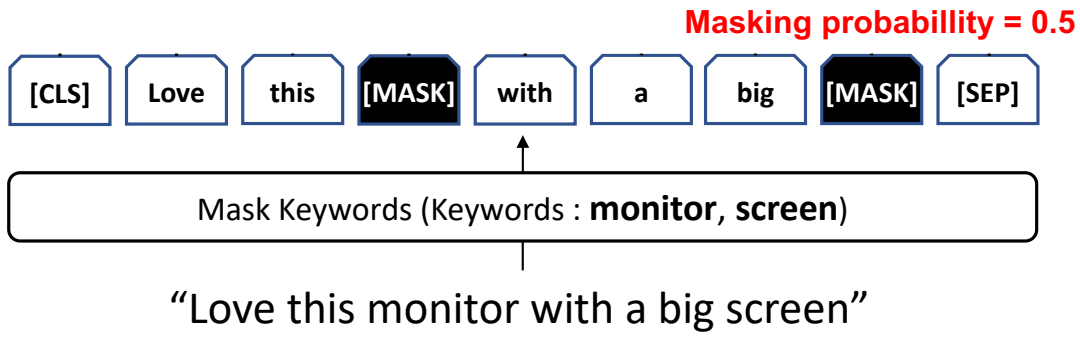
MKR: Following context around a few keywords

- MKR forces model to guess keywords by using context around it



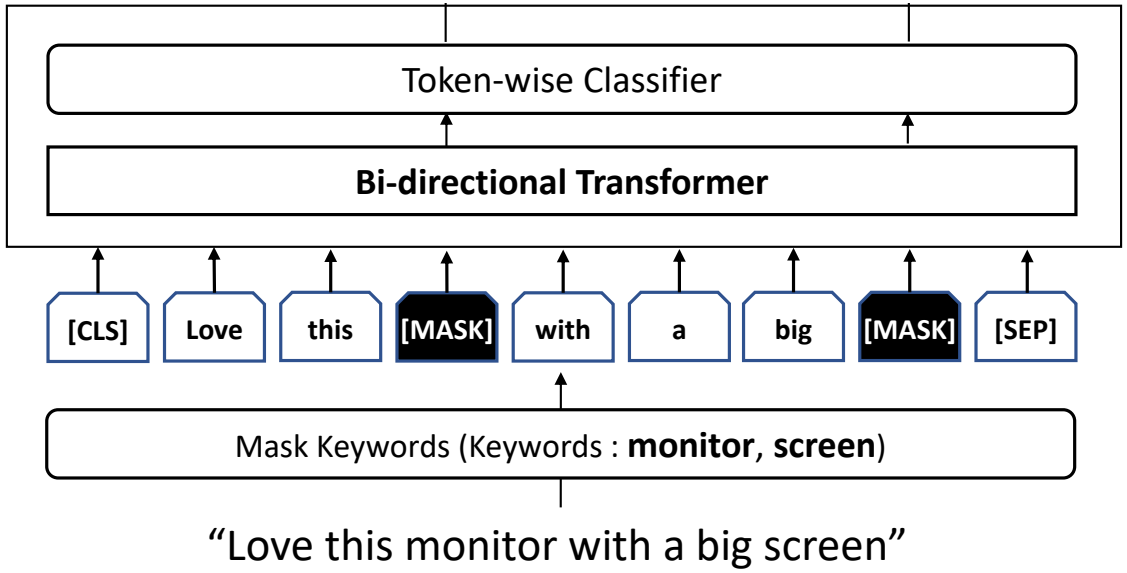
MKR: Following context around a few keywords

- First, MKR masks keywords in the input text
  - Since Masking all keywords might harm context severely, we randomly mask half of the keywords



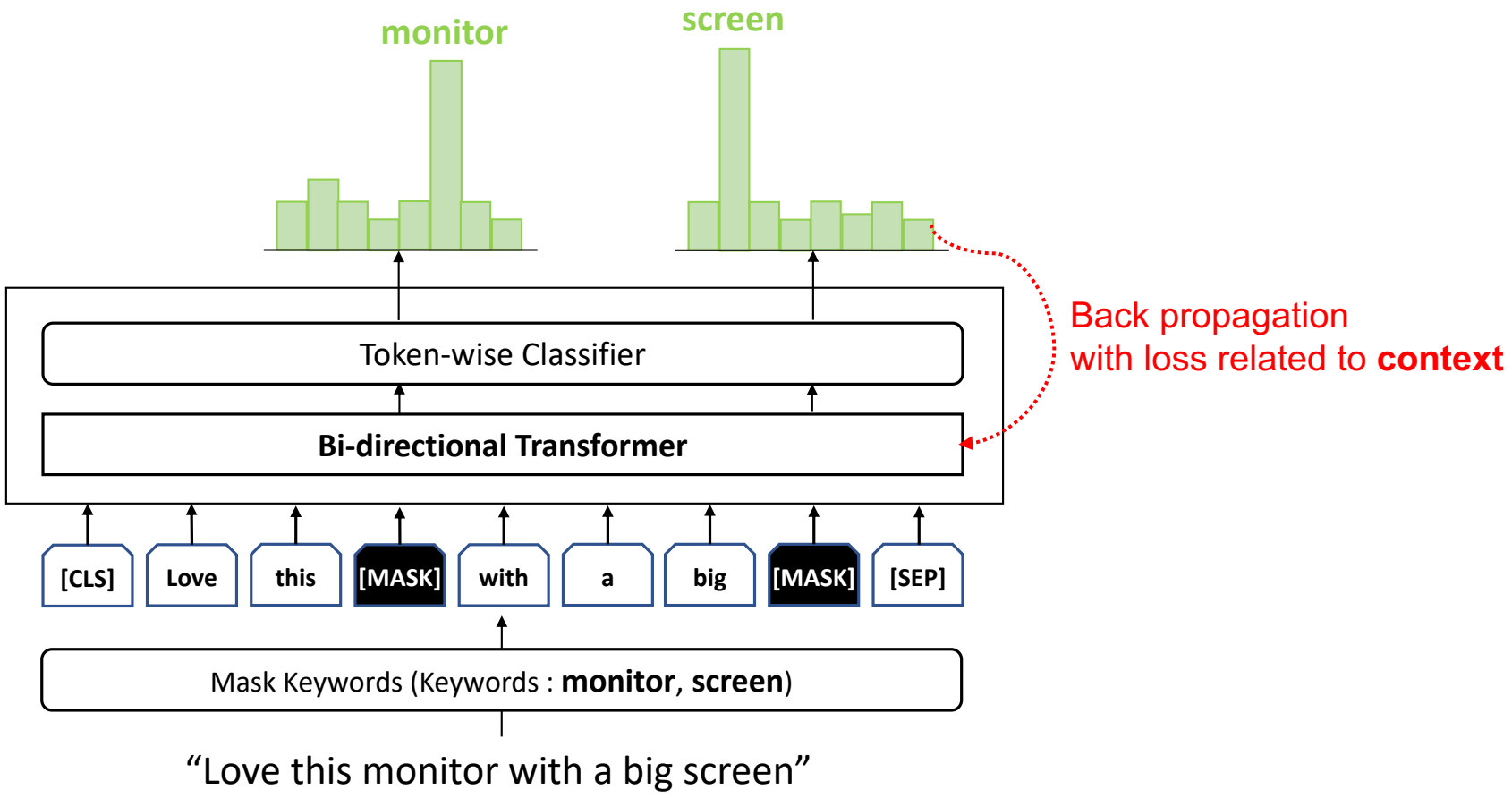
MKR: Following context around a few keywords

- We put self-supervision head following BERT structure (Token-wise Classifier)
  - Token-wise classifier yields logits with a size of **total keywords**



MKR: Following context around a few keywords

- Then, MKR forces the model to guess the keywords by using context around it
  - MLM is known to effective for making models see the context



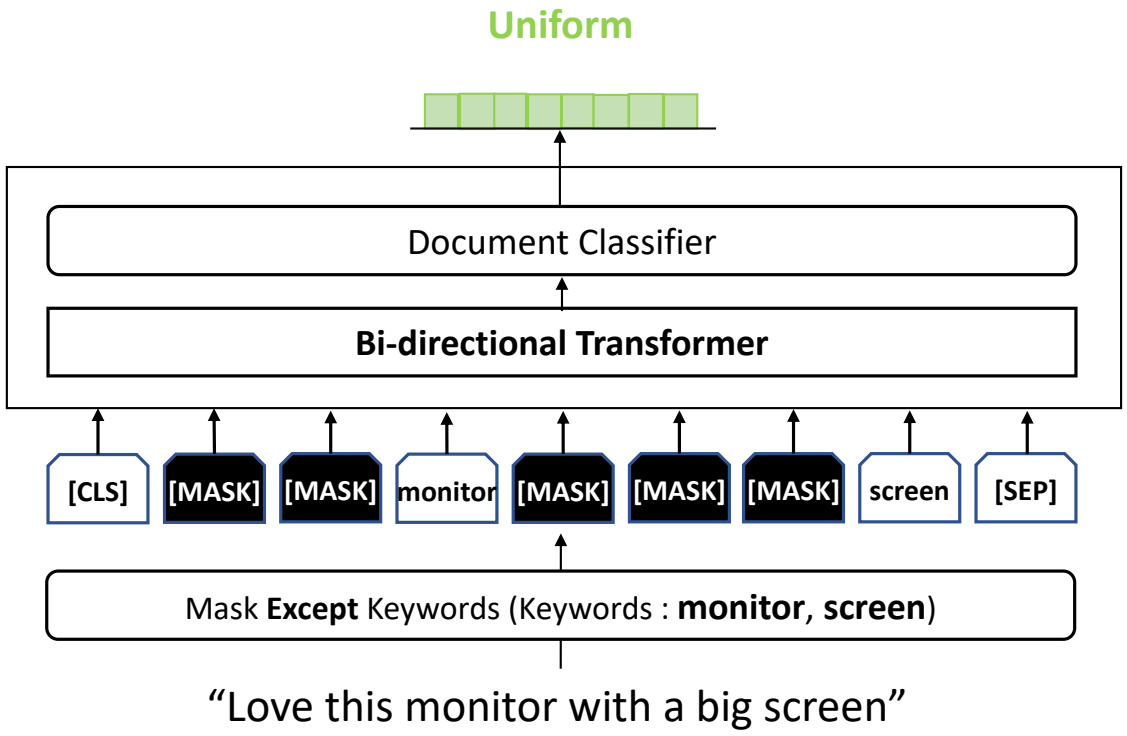
MER: Making OODs containing keywords

- But still document classifier can rely on the frequency of keywords

⇨ We figure out **Masked Entropy Regularization (MER)**

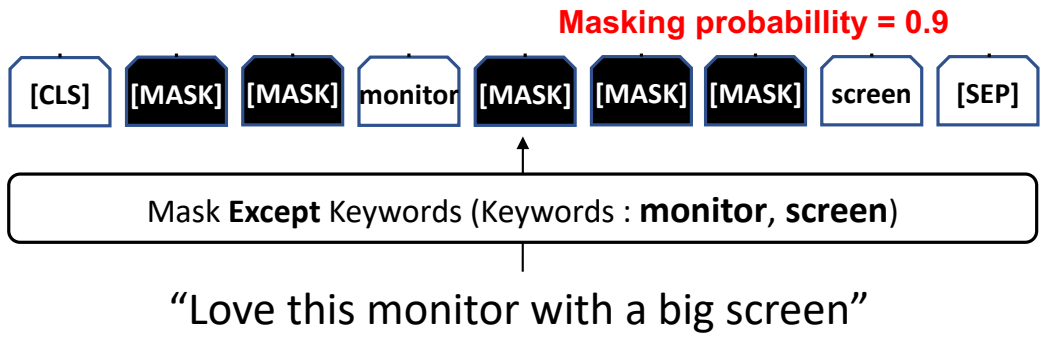
MER: Making OODs containing keywords

- MER directly regularizes document classifier, which is used at inference stage



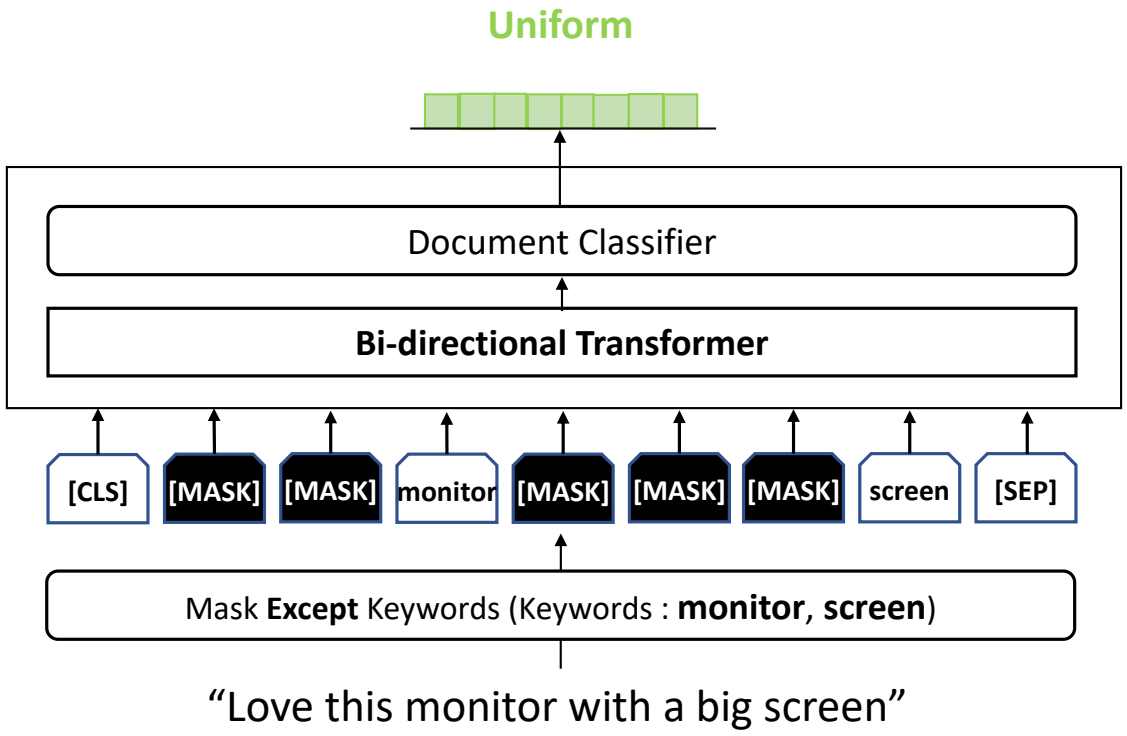
MER: Making OODs containing keywords

- First, MER masks every token **except** keywords
  - For making a meaningless sentence, MER masks tokens with a probability 90%, except keywords
- We can get a meaningless sentence but contains keywords



MER: Making OODs containing keywords

- MER regularizes document classifier to yield uniform distribution for inputs
- Model learns the **existence of keywords** is **not** a standard for classification



Keyword selection method

- We select keyword in two ways:
  - Frequency-Based
  - Attention-Based

Frequency-based keyword selection

- We obtain TF-IDF score for each token for each class in the dataset
- We use the highest value among each TF-IDF per class, as a score:

$$s^{\texttt{freq}}(t) = \max_{c \in \{1,\ldots,C\}} \texttt{tf}(t, \mathbf{X}_c) \cdot \texttt{idf}(t, \mathbf{D})$$

Attention-based keyword selection

- We score every token by averaging its attention weight to [CLS] at the final layer
- Let

  a = $[a_1, a_2 \cdots, a_T]$ be attention values of document embedding

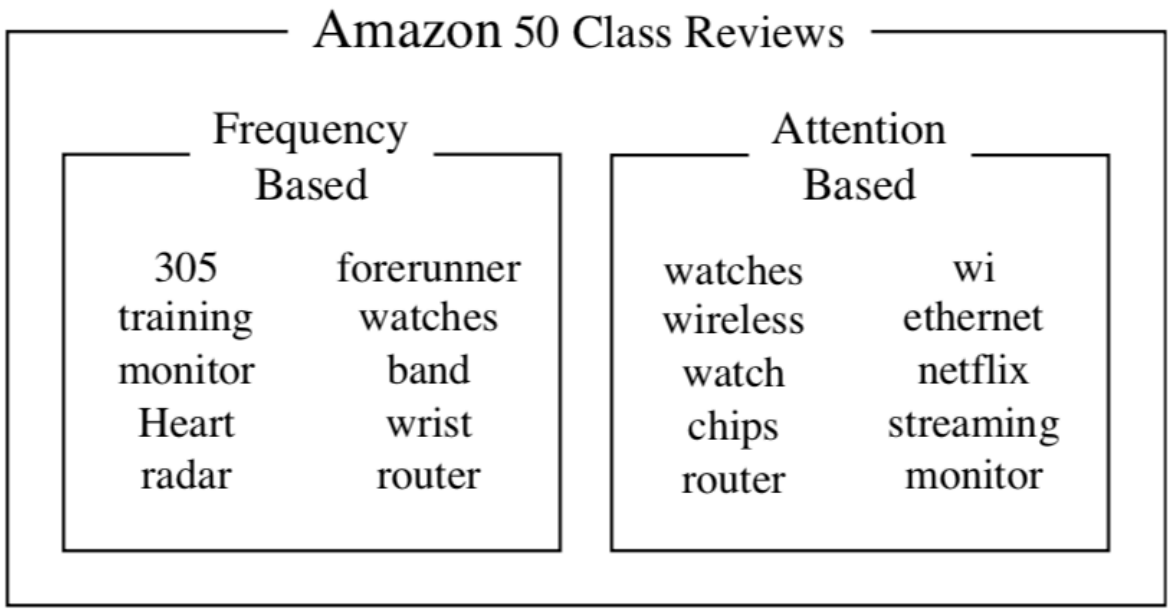
  $n_{t,x}$ is the number of token *t* in document **x**
- We obtain attention score of token *t* as following:

Indicator Function

$$s^{\texttt{attn}}(t) = \sum_{(\mathbf{x},y)\in\mathcal{D}} \frac{1}{n_{t,\mathbf{x}}} \sum_{i\in\{1,\ldots,T\}} \mathbb{I}(t_i = t) \cdot \frac{a_i}{\|\mathbf{a}\|}$$

Comparison between keyword selection method

- We compare keywords selected by two different methods
- Both methods select meaningful keywords
- Keywords selected by the attention-based method is more generalizable



Experiments: OOD detection

- MASKER shows consistent improvement on AUROC(%), compared to vanilla models

ID	OOD	OC-SVM	DOC	Vanilla / Residual / MASKER		
				BERT	RoBERTa	ALBERT
Newsgroup	Amazon	62.1	84.1	85.4/86.7/**87.0**(+1.9%)	85.3/85.9/**87.2**(+2.3%)	86.7/85.4/**89.4**(+3.1%)
	Reuters	53.9	60.0	91.8/93.0/**97.7**(+6.4%)	93.1/92.1/**93.9**(+0.8%)	93.3/92.0/**94.7**(+1.6%)
	IMDB	59.8	88.6	94.6/95.7/**98.5**(+4.1%)	95.2/95.9/**97.7**(+2.7%)	94.5/92.6/**96.6**(+2.2%)
	SST-2	63.0	88.1	87.0/97.0/**98.6**(+13.3%)	94.7/94.9/**98.2**(+3.6%)	95.8/95.6/**98.1**(+2.4%)
	Fine Food	62.8	81.3	85.3/87.3/**93.4**(+9.5%)	88.7/89.4/**92.9**(+4.7%)	77.6/86.6/**91.6**(+18.1%)
Amazon	Newsgroup	61.3	81.3	84.8/83.9/**87.2**(+2.8%)	87.9/86.6/**91.0**(+3.5%)	87.3/85.0/**88.4**(+1.3%)
	Reuters	55.5	79.8	89.7/89.7/**93.5**(+4.2%)	92.3/92.6/**93.6**(+1.4%)	93.1/93.5/**94.5**(+1.5%)
	IMDB	66.2	89.6	93.3/92.8/**95.2**(+2.0%)	90.1/87.0/**93.3**(+3.5%)	89.9/88.6/**95.6**(+6.4%)
	SST-2	60.9	91.5	93.0/88.9/**95.6**(+2.8%)	92.4/94.8/**96.4**(+4.4%)	93.4/91.9/**96.9**(+3.7%)
	Fine Food	51.1	66.8	78.5/77.7/**84.9**(+8.2%)	74.9/80.0/**80.7**(+7.8%)	82.6/86.3/**87.3**(+5.7%)

Experiments: Cross-domain generalization

- MASKER shows consistent improvement for various cross-domain generalization task

Train	Test	OpenMax	DOC	Vanilla / Residual / MASKER BERT	RoBERTa	ALBERT
IMDB	IMDB	87.7	88.0	93.5/92.8/93.5	95.3/94.0/95.6	91.6/91.4/90.8
	SST-2	79.6	77.9	85.9/86.9/**88.1**(+2.6%)	89.7/90.2/**91.8**(+2.3%)	89.8/89.0/**89.9**(+0.1%)
	Fine Food	75.4	78.3	92.9/92.5/**93.6**(+0.8%)	92.6/87.7/**93.0**(+0.4%)	87.1/87.8/**92.1**(+5.7%)
SST-2	SST-2	82.9	83.1	92.5/90.3/92.3	94.5/92.0/94.3	91.9/90.9/91.5
	IMDB	75.3	76.9	85.7/86.2/**88.4**(+3.2%)	86.0/85.7/**87.3**(+1.5%)	83.0/83.0/**83.8**(+1.0%)
	Fine Food	62.2	64.5	87.6/88.2/**89.2**(+1.8%)	86.5/87.8/**89.6**(+3.6%)	79.0/80.6/**84.9**(+7.5%)
Fine Food	Fine Food	93.6	93.3	96.5/94.8/96.5	96.9/95.7/97.1	95.5/95.4/96.6
	IMDB	67.5	67.3	77.3/81.1/**85.6**(+10.7%)	84.1/84.6/**86.6**(+2.9%)	74.7/80.4/**84.8**(+13.5%)
	SST-2	61.9	62.0	67.6/67.8/**80.0**(+18.3%)	78.5/80.1/**83.8**(+6.8%)	71.7/73.2/**83.3**(+16.2%)

(a) Sentiment analysis

Model	Telephone (ID)	Telephone Letters (OOD)	Telephone Face-to-Face (OOD)
BERT	80.5	77.4	77.5
+MASKER	80.2	**80.4** (+3.9%)	**78.5** (+1.3%)
RoBERTa	84.8	83.6	83.5
+MASKER	86.0	**85.9** (+2.8%)	**83.8** (+0.4%)
ALBERT	82.9	82.8	80.9
+MASKER	82.5	**84.0** (+1.5%)	**86.8** (+7.3%)

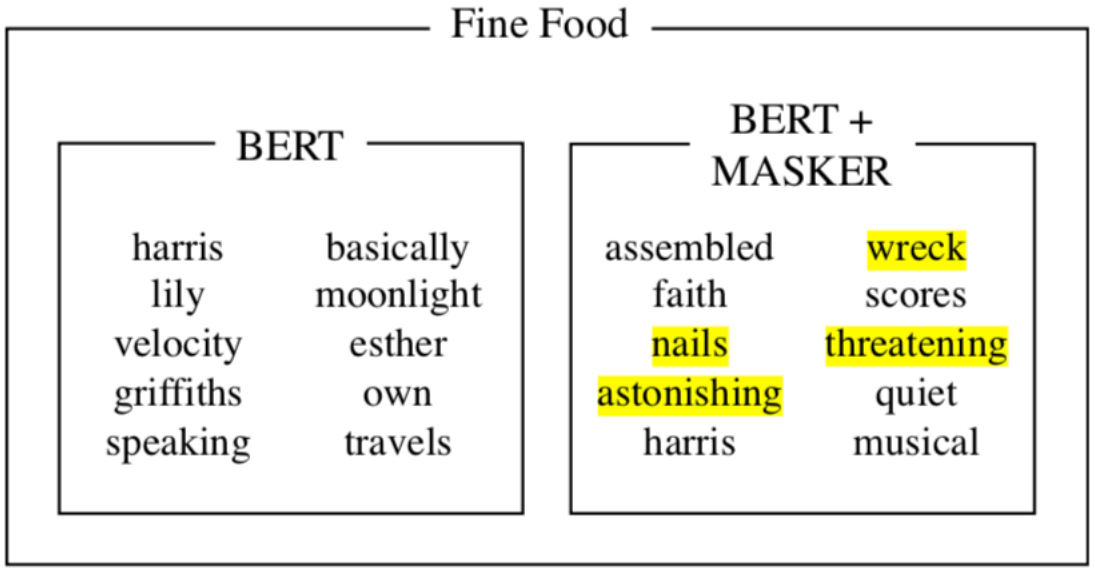
(b) Natural language inference

MSRvid (ID)	MSRvid Images (OOD)	MSRvid Headlines (OOD)	Images (ID)	Images MSRvid (OOD)	Images Headlines (OOD)
91.5	82.0	61.7	88.0	89.7	73.9
91.2	**84.3** (+2.8%)	**66.7** (+8.1%)	88.1	**91.6** (+2.1%)	**75.3** (+1.9%)
94.2	**88.0**	80.3	91.8	92.9	84.1
93.7	**88.0** (+0.0%)	**84.0** (+4.6%)	91.3	**94.1** (+1.3%)	**85.3** (+1.4%)
92.6	81.2	60.6	90.4	90.9	69.8
93.3	**82.6** (+1.7%)	**68.8** (+13.5%)	90.5	**92.0** (+1.2%)	**78.4** (+12.3%)

(c) Semantic textual similarity

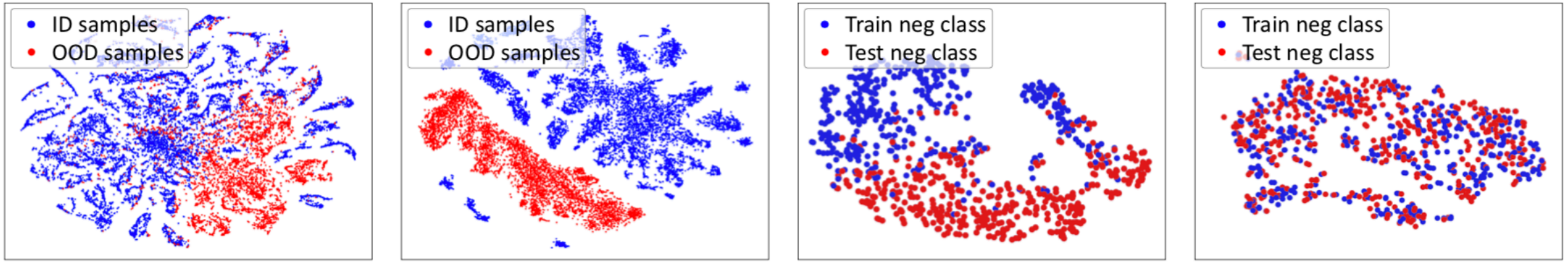
Experiments: Cross-domain generalization

- We compare selected keywords for each training method
- BERT+MASKER picks more topic-related(highlighted) keywords than vanilla BERT



Experiments: t-SNE plot

- We can see MASKER
  - Distinguishes OOD data clearly compared to the vanilla model
  - Better entangles the same classes in training and test datasets



(a) BERT (b) MASKER (ours) (c) BERT (d) MASKER (ours)

Thank you for your attention ☺

Paper: arxiv.org/abs/2012.09392

Code: https://github.com/alinlab/MASKER